

Polyominoes with a Saturated Cell

Each figure shows one example free polyomino of n cells that contains a saturated cell – a cell all four of whose edge-neighbours are present, so it sits fully surrounded in the interior. Cells are filled teal; the saturated cell is the interior one with no exposed edge (the centre of a plus shape). The value $a(n)$ is the total number of free polyominoes of n cells with at least one saturated cell (the figure is just one representative). The first such shape is the plus-pentomino at $n = 5$.

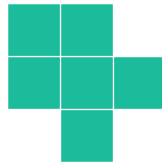
$$a(5) = 1, a(6) = 2, a(7) = 9, a(8) = 36, a(9) = 157, a(10) = 667$$

Computed by Peter Exley, Jun 2026

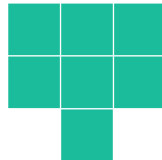
$$a(5) = 1 \text{ [PROVED] example, } 3 \times 3 \text{ bbox}$$



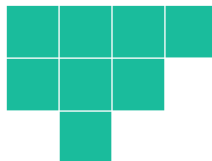
$$a(6) = 2 \text{ [PROVED] example, } 3 \times 3 \text{ bbox}$$



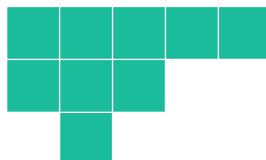
$$a(7) = 9 \text{ [PROVED] example, } 3 \times 3 \text{ bbox}$$



$$a(8) = 36 \text{ [PROVED] example, } 3 \times 4 \text{ bbox}$$



$$a(9) = 157 \text{ [PROVED] example, } 3 \times 5 \text{ bbox}$$



$$a(10) = 667 \text{ [PROVED] example, } 3 \times 6 \text{ bbox}$$

